Maryam Alipourhajiagha

Canada, Montreal (open to relocate) | maryamalipour9877@gmail.com | +14385290440
maryamalipour.io | linkedin.com/in/maryam-alipourhajiagha | github.com/maryamalipourh | google scholar

Summary

AI/ML Research and engineer specializing in the design and development of reliable, uncertainty-aware deep learning systems. Proven expertise in architecting probabilistic generative models and high-dimensional computer vision pipelines, with a deep foundation in mathematics. Expert in the end-to-end ML lifecycle, using PyTorch and SLURM-based multi-GPU workflows. Researcher with hands-on experience fine-tuning LLMs and optimizing models for explainability and reliability.

Education

MASc in Applied Mathematics @ *Polytechnique Montreal* 01/2024 – 04/2026
- Courses: Representation Learning, Statistical ML, Machine Learning Applied to Climate Change, Deepl Learning in NLP (GPA 3.91)
- TA: Artificial Intelligence: Probabilistic and Learning Techniques, Fundamentals of ML, Unsupervised Learning and Time Series

BSc in Applied Mathematics, minor in Computer Science @ *AmirKabir University of Technology* 09/2017 – 07/2022
- TA: Numerical Computations, Numerical Analysis, Fundamentals of Programming in C

Publications

Explainable Generative Models for Climate Downscaling @ *TBD* in prep
- M Alipourhajiagha, J Carreau, B Gauvin

A Probabilistic U-Net Approach to Downscaling Climate Simulations @ *NeurIPS 2025 (AI4Science)* 10/2025
- M Alipourhajiagha, PL Lemaire, Y Diouane, J Carreau

Work Experience

Machine Learning Research Intern @ *Ouranos* 09/2025 – 04/2026
- Conducting applied AI research on probabilistic deep models to improve their interpretability and scientific analysis workflows.
- Extending generative models to learn from ensembles of high-dimensional simulations to improve the learned representations.
- Capturing uncertainty in models using ensemble-based training, probabilistic, and distribution-aware evaluation techniques.

Graduate Research Assistant @ *Mila - Quebec Artificial Intelligence Institute* 01/2024 – 04/2026
- Worked on deep learning for climate downscaling, focusing on generating high-resolution data from coarse climate simulations.
- Developed a probabilistic model (NeurIPS'25@AI4Science) producing plausible high-resolution outputs and capture uncertainty.
- Studied objectives to balance predictive accuracy, realistic spatial detail, and extreme-event behavior in probabilistic models

Teaching Assistant @ *CS50x Iran* 08/2020 – 04/2022
- Contributed to CS50x, Harvard's introductory computer science course, by adopting the material for Persian-speaking audience.
- Customized course content, supported TA sessions for 2020-2022 cohorts, and reviewed assignments and held Q&A sessions.

Technical Skills

Programming & Scripting: Python, C, Java, SQL, Bash, HTML/CSS
Libraries & Frameworks: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Hugging Face (Transformers, Datasets, Diffusers), WandB, TensorBoard, Matplotlib, Seaborn, Plotly
Core AI/ML Knowledge: Supervised and Unsupervised Learning, Deep Learning (CNNs, RNNs, Transformers), Probabilistic Modeling, Self-Supervised and Contrastive Learning, Transfer Learning and Domain Adaptation, Generative Models (VAEs, GANs, Diffusion Models), Large Language Models (LLMs), Evaluation Metrics and Benchmarking, Experiment Design and Ablation Studies, Model Distillation, Interpretability and Explainability, Multi-GPU Training (SLURM-based workflows on Compute Canada)
Tools: Git, Docker, MySQL, Jupyter, Linux, LaTeX, AI-assisted Coding (Claude, Cursor)

Projects

Emotion Recognition from Conversational Context @ *GitHub* 04/2025
- Developed hybrid models (e.g., SS-BED and CNN–BiGRU) for emotion recognition in text to balance accuracy, speed, and resource efficiency, practical alternatives to transformer models in resource-constrained applications.

LitGPT @ *GitHub* 03/2025
- Implemented parameter-efficient instruction tuning by applying LoRA to TinyLlama and Pythia on LIMA and benchmarking zero-shot performance on HellaSwag, applied an improved LoRA variant and compared results.

Evaluation of Transformer-based Models @ *GitHub* 02/2025
- Finetuning and evaluating transformer-based models like BERT, DistilBERT, GPT-2, RoBERTa, ELECTRA on diverse datasets like ISEAR, Diar AI, and GoEmotions to perform emotion classification on texts.